

Early-Stopping Activation Steering for Factual Preference Alignment in Vietnamese Medical Language Models

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Abstract. Small Language Models (SLMs) offer potential for privacy-preserving clinical decision support but frequently exhibit domain-specific hallucinations in resource-scarce languages like Vietnamese—such as miscalculating pediatric dosages or violating pregnancy contraindications. We propose Early-Stopping Activation Steering, a framework that dynamically shifts model outputs toward factual preference at inference time with no weight updates. Using our newly constructed ViHaluEval-Medical benchmark (14,674 records from the Vietnamese National Drug Formulary, 2nd Edition, 2018), we evaluate steering interventions on Qwen2.5-7B-Instruct at Layer 8 with a linear decay mechanism restricting vector injection to the first $K=16$ decoding steps ($t \in [1, 16]$, influencing output tokens $y_2 \dots y_{17}$). Under high-cap natural completion (`max_new_tokens=800`), Hard Cutoff ($K=16$) achieves 70.60% reference-preference accuracy (+5.20 pp over baseline, Holm-adjusted $p_{\text{adj}} = 0.0026$) with 100% EOS hit rate. Under bounded-generation stress testing (`max_new_tokens=200`), Linear Decay steering ($K=16$) increases accuracy from 73.00% to 77.20% (+4.20 pp, McNemar $p = 0.0055$), outperforming non-oracle BM25 RAG by +8.60 pp with zero context overhead. Placebo controls ($N=100$ random vectors, $55.82\% \pm 4.15\%$) confirm directional specificity at $+5.15\sigma$ above random perturbation.

Keywords: Activation steering · Hallucination mitigation · Clinical language models · Vietnamese NLP · Inference-time intervention · Representation engineering.

1 Introduction

Large Language Models (LLMs) have demonstrated capabilities across healthcare tasks [3]. However, in specialized domains requiring high factual precision—such as clinical pharmacology and prescription guidance in non-English languages—standard models suffer from domain-specific hallucinations [7]. In Vietnamese clinical settings, these hallucinations range from benign stylistic errors to dangerous medical contradictions, such as recommending contraindicated medications during pregnancy or miscalculating weight-based pediatric dosages.

Traditional mitigation strategies rely on post-hoc supervised fine-tuning (SFT) [10] or Retrieval-Augmented Generation (RAG) [5]. Although RAG provides external knowledge contexts, models may ignore retrieved evidence when internal activation pathways favor non-truthful generation [11], additionally incurring

heavy context window overheads and retrieval latencies. Supervised fine-tuning demands substantial computational resources and risks catastrophic forgetting of general abilities [12].

Recently, Representation Engineering (RepE) [1] and Inference-Time Intervention (ITI) [2] have emerged as lightweight, non-destructive alternatives. By extracting a truthfulness steering vector v_{steer} from hidden activation contrasts and injecting it into intermediate layers during generation, activation steering shifts model outputs toward factual accuracy without modifying underlying weights. Related approaches such as Activation Addition (ActAdd) [8] and contrastive activation addition [9] have demonstrated the viability of linear steering for behavioral control.

However, continuous activation steering ($K = \infty$) induces persistent hidden-state norm elevation over long output sequences, associated with cumulative activation displacement and text repetition. In this paper, we present **Early-Stopping Activation Steering**, a temporal decay mechanism restricting vector injection to the first $K = 16$ decoding steps ($t \in [1, 16]$), influencing tokens $y_2 \dots y_{17}$.

Our main contributions are: (i) **ViHaluEval-Medical Benchmark:** A benchmark of 14,674 expert-structured Vietnamese clinical Q&A pairs derived from the official Vietnamese National Drug Formulary (2nd Edition, 2018) [6], featuring a question-disjoint test split ($N_{\text{test}} = 500$). (ii) **Dual-Evaluation Paradigm:** Comprehensive evaluation covering high-cap natural completion ($\text{max_new_tokens}=800$, EOS Hit = 99.80%–100.00%) and bounded-generation stress testing ($\text{max_new_tokens}=200$). (iii) **Empirical Activation-Norm Dynamics:** Hidden-state activation tracking ($\|\hat{h}_g^{(t)}\|_2$) demonstrating persistent norm elevation under continuous steering (56.34 vs 53.39 baseline at $t = 50$, $\Delta = +2.95$) and early-stopping magnitude relaxation back to baseline equilibrium (53.39 ± 0.64 sample mean $\pm$ SD, $\Delta = 0.00$) under fixed teacher-forcing context. (iv) **Directional Specificity & Multi-Random Placebo:** Intervention along negative steering direction ($-v_{\text{steer}}$, achieving 62.80% accuracy), 50% label-permutation placebo steering (v_{shuf} , 68.60% accuracy, $N_{\text{flipped}} = 5,198/10,272$), and distributional validation across $N = 100$ independent random vectors ($55.82\% \pm 4.15\%$ SD) confirming $+v_{\text{steer}}$ (77.20%) operates at a descriptive $+5.15\sigma$ distance above random direction perturbation ($p = 1/101 = 0.0099 < 0.01$), alongside a non-oracle BM25 RAG baseline (68.60% accuracy, Recall@1 = 19.20%). (v) **Reproducible Statistical Validation:** Exact paired 2×2 binomial McNemar testing ($p = 0.0055$, 95% Bootstrap CI [+1.40 pp, +7.00 pp] for bounded stress testing; exact McNemar $p = 0.00086$, 95% Bootstrap CI [+2.20 pp, +8.20 pp] for natural completion) and Wilcoxon signed-rank testing ($p = 0.017$) confirming reference-preference gains. (vi) **Controlled Factorial Ablation:** Controlled 3×3 equal-initial- α_0 matrix alongside a separately specified nominal matched cumulative dose control (α_{match}).

The remainder of this paper is organized as follows: Section 2 reviews related work, Section 3 presents methodology, Section 4 reports experimental results, Section 5 discusses limitations, and Section 6 concludes.

2 Related Work

2.1 Hallucination in Medical Language Models

LLM hallucinations in clinical domains pose unique risks compared to general-purpose applications. Ji et al. [7] survey hallucination phenomena, categorizing them into intrinsic and extrinsic types. In medical settings, Singhal et al. [3] showed that while large models encode clinical knowledge, they remain prone to factual errors in drug interactions, dosage calculations, and contraindications. The HaluEval benchmark [13] introduced evaluation across diverse domains, inspiring our domain-specific ViHaluEval-Medical benchmark for Vietnamese clinical texts. Lin et al. [16] proposed TruthfulQA for measuring model truthfulness, establishing evaluation protocols that informed our contrastive design.

2.2 Inference-Time Intervention and Representation Engineering

Rather than modifying model weights through parameter-efficient fine-tuning [10], inference-time methods intervene directly in internal representations during generation. Li et al. [2] proposed ITI, identifying truthful directions in attention head outputs. Zou et al. [1] generalized this to Representation Engineering (RepE), demonstrating that high-level concepts can be extracted and controlled via linear subspace manipulation. Turner et al. [8] introduced ActAdd, showing that simple vector addition to residual streams steers model behavior. Rinsky et al. [9] extended this with contrastive activation addition. Our work builds upon these foundations but addresses the limitation of continuous steering over long sequences by introducing temporal decay.

2.3 Evaluation Metrics for Clinical Text Generation

Traditional n-gram overlap metrics like ROUGE [14] provide lexical similarity scores but fail to capture semantic equivalence in multilingual clinical settings. Zhang et al. [15] proposed BERTScore, which computes token-level cosine similarities using contextual BERT embeddings, offering evaluation robust to lexical variation. We adopt a BERTScore-based reference-preference criterion as a proxy for factual preference alignment.

3 Methodology

3.1 ViHaluEval-Medical Benchmark Construction

The ViHaluEval-Medical benchmark comprises 14,674 expert-structured Vietnamese clinical Q&A pairs constructed directly from the official Vietnamese National Drug Formulary (Được thư Quốc gia Việt Nam, 2nd Edition, 2018) [6]. The primary source text spans 14.1 million characters across 1,026 comprehensive drug monographs (`duoc_thu_2018_cleaned.txt`). To construct contrastive pairs, raw monographs were segmented into 1,000-character context chunks, maintaining direct source-to-record traceability where each chunk maps to generated clinical question records (q_i). A high-throughput vLLM inference engine powered by `Qwen2.5-7B-Instruct-AWQ` (batch size 64, temperature = 0.7, top- p = 0.9, max tokens = 650) executed a structured expert prompt. For each context chunk,

the engine generated distinct clinical question records (q_i), each paired with a 100% factually accurate reference answer (y_i^+) and a single category-targeted hallucinated counter-answer y_i^- (`hallucinated_answer`, drawn from dosage modifications $y_{i,1}^-$, pregnancy safety contradictions $y_{i,2}^-$, or drug-drug interactions $y_{i,3}^-$). Each record in the final manifest (`vietnamese_medical_halueval_15k_specialized.json`) represents an unambiguous contrastive triplet (q_i, y_i^+, y_i^-), ensuring contrastive vector estimation ($\mu^+ - \mu^-$) and BERTScore RefPref evaluation operate on single defined counter-answer pairs. The full dataset spans three clinical categories: Special Dosage Modifications ($N = 5,000$, 34.08%), Contradictory Pregnancy Safety ($N = 4,885$, 33.29%), and Misleading Drug Interactions ($N = 4,789$, 32.64%), strictly summing to $N = 14,674$ total records.

All candidate Q&A pairs underwent manual adjudication by 3 licensed Vietnamese clinical pharmacists (≥ 5 years clinical experience in hospital settings). Annotators cross-referenced each generated record against source drug monograph passages to verify 100% medical accuracy of reference answer y_i^+ and confirm counter-answer y_i^- represented a plausible clinical hallucination, achieving high inter-annotator agreement (mean pairwise Cohen’s $\kappa = 0.91$; Fleiss’ $\kappa = 0.89$). Benchmark data is partitioned at the active pharmaceutical ingredient (API) level into a development split ($N_{\text{dev}} = 12,495$ pairs, 85.15%: comprising 70.0% training split $N_{\text{train}} = 10,272$ for activation mean estimation and 15.15% validation split $N_{\text{val}} = 2,223$ for hyperparameter tuning). Contrastive steering vector v_{steer} is estimated strictly using training split ($N_{\text{train}} = 10,272$ pairs) to compute baseline activations μ^+ and μ^- ; all directional control vectors (v_{shuf} , covariance-matched) are constructed on the same $N_{\text{train}} = 10,272$ pool. Hyperparameter selection (layer l , cutoff K , scale α_0) is performed exclusively on a $N_{\text{eval}} = 500$ validation subset drawn from $N_{\text{val}} = 2,223$, keeping primary test set ($N_{\text{test}} = 500$) strictly held-out. Held-out test partition ($N_{\text{test_partition}} = 2,179$ pairs, 14.85%) is reserved for the evaluation subset of $N_{\text{test}} = 500$ questions sampled using stratified random sampling with fixed seed 42, maintaining balanced category representation: Pregnancy Safety ($N = 165$), Drug Interactions ($N = 160$), and Special Dosage ($N = 175$). Numerical difference between 14,674 Q&A pairs and 14,576 formulary passages arises because 98 multi-indication monographs yielded 2 distinct clinical question pairs.

3.2 Contrastive Vector Estimation & Directional Controls

Given a clinical prompt x_i , let y_i^+ denote the ground-truth medical answer and y_i^- denote an intentionally hallucinated counter-answer. We collect internal hidden activations $h_l(x_i, y_i)$ at layer l using a forward hook attached to the final token position of the concatenated sequence:

$$h_l(x_i, y_i) = \text{LayerOutput}_l(x_i \oplus y_i)[: , -1, :] \quad (1)$$

The truthfulness steering vector $v_{\text{steer}}^{(l)}$ is computed as normalized mean difference between positive and negative activation distributions:

$$v_{\text{steer}}^{(l)} = \frac{\mu_l^+ - \mu_l^-}{\|\mu_l^+ - \mu_l^-\|_2} \quad (2)$$

where $\mu_l^+ = \frac{1}{N} \sum_{i=1}^N h_l(x_i, y_i^+)$ and $\mu_l^- = \frac{1}{N} \sum_{i=1}^N h_l(x_i, y_i^-)$.

To provide rigorous directional evidence against generic residual perturbation, we evaluate four baseline control conditions: (i) **Negative Steering** ($-v_{\text{steer}}$): Steering along anti-truthful direction using $\hat{h}_8^{(t)} = h_8^{(t)} - |\alpha(t)| \cdot v_{\text{steer}}$ ($\alpha_0 = -18.0$) to test if intervention symmetrically degrades factual performance. (ii) **Full Label-Permutation** (v_{shuf}): Constructed by randomly swapping prompt labels ($y_i^+ \leftrightarrow y_i^-$) across $N_{\text{flipped}} = 5,198/10,272$ training pairs (50.60% permutation balance) on $N_{\text{train}} = 10,272$ pool, achieving 68.60% RefPref (343/500) and BERTScore F1 0.6845. (iii) **Covariance-Matched Distribution** ($N = 20$ **vectors, seeds 3000–3019**): Sampled as $v_{\text{cov}}^{(r)} = \frac{Lz_r}{\|Lz_r\|_2}$ using Ledoit-Wolf shrinkage matrix factorization $\Sigma_{\text{data}} = LL^T$ from residual difference vectors $d_i = h_i^+ - h_i^-$, evaluating across 10,000 test passes ($\pm 0.95\%$ sample SD). (iv) **Isotropic Distribution** ($N = 100$ **vectors**): $\{v_{\text{rand}}^{(r)}\}_{r=1}^{100} = z_r / \|z_r\|_2$ for $z_r \sim \mathcal{N}(0, I_{3584})$.

3.3 Early-Stopping Steering Taxonomy & Activation-Norm Dynamics

Under our proposed **Early-Stopping Activation Steering** framework, residual stream activations $h_l^{(t)}$ at Layer 8 (decoder block index 8, 9th block of Qwen2.5-7B’s 28-layer architecture) are modified during autoregressive generation ($t \geq 1$) according to:

$$\hat{h}_l^{(t)} = h_l^{(t)} + \alpha(t) \cdot v_{\text{steer}}^{(l)} \quad (3)$$

We evaluate three distinct temporal intervention schedules under fixed α_0 : (i) **Linear Decay Schedule** ($K = 16$): Coefficient decreases linearly across the first K decoding steps ($t \in [1, K]$), remaining strictly non-zero at step $t = K$ ($\alpha_{\text{decay}}(K) = \frac{\alpha_0}{K}$) and deactivating ($\alpha = 0$) for subsequent decoding steps $t > K$:

$$\alpha_{\text{decay}}(t) = \begin{cases} \alpha_0 (1 - \frac{t-1}{K}), & 1 \leq t \leq K \\ 0, & t > K \end{cases} \quad (4)$$

(ii) **Hard Cutoff Schedule** ($K = 16$): Step-function hook deactivation after K post-prefill decoding steps ($t \in [1, K]$):

$$\alpha_{\text{cutoff}}(t) = \begin{cases} \alpha_0, & 1 \leq t \leq K \\ 0, & t > K \end{cases} \quad (5)$$

(iii) **Continuous Steering Schedule** ($K = \infty$): Unbounded steering applied across all generated decoding steps ($t \geq 1$): $\alpha_{\text{continuous}}(t) = \alpha_0$ for all $1 \leq t \leq T_{\text{gen}} - 1$.

Cumulative intervention dose depends on temporal window K , initial strength α_0 , and total generated sequence length T_{gen} post-prefill. Because unsteered prompt prefill precedes decoding ($t = 0$), generation under `max_new_tokens=200` executes up to 199 actual decoding steps ($t \in [1, 199]$). For Continuous Steering ($K = \infty, \alpha_0 = 18.0$), vector injection applies across $T_{\text{gen}} - 1$ decoding steps, yielding executed dose $D_{\text{continuous, executed}} = (T_{\text{gen}} - 1) \cdot |\alpha_0|$; under full 200-token generation (199 decoding steps), executed dose reaches $199 \times 18.0 = 3,582.0$. For early-stopping schedules ($K = 16, \alpha_0 = 18.0$), executed dose is $D_{\text{cutoff}} = K \cdot |\alpha_0| = 16 \times 18.0 = 288.0$ under Hard Cutoff, and $D_{\text{decay}} =$

$\sum_{t=1}^K |\alpha_0| \left(1 - \frac{t-1}{K}\right) = \frac{(K+1)}{2} |\alpha_0| = 153.0$ under Linear Decay, provided output length $T_{\text{gen}} \geq 17$. For Nominal Matched Cumulative D_1 Dose Control ($\alpha_0 > 0$), constant per-token coefficient $\alpha_{\text{match}} = \text{sgn}(\alpha_0) \frac{D_{\text{decay}}}{T} = \text{sgn}(\alpha_0) \frac{K+1}{2T} |\alpha_0|$ derived under nominal sequence length $T = 200$, yielding $\alpha_{\text{match}} = 0.0425\alpha_0$ (applied coefficients 0.6375, 0.7650, and 0.8500 for source values $\alpha_0 = 15, 18, 20$). For Nominal Matched-Dose Control, actual executed cumulative dose scales with total generated sequence length T_{gen} post-prefill according to:

$$D_{\text{executed}} = (T_{\text{gen}} - 1) \cdot |\alpha_{\text{match}}| \quad (6)$$

Under full 200-token generation ($T_{\text{gen}} = 200$, 199 decoding steps post-prefill), executed dose reaches $(200 - 1) \cdot 0.7650 = 199 \cdot 0.7650 = 152.235$ (within 0.5% of 153.0 target). If natural EOS termination occurs prior to $T_{\text{gen}} = 200$, executed dose scales proportionally with decoding length. We retain the terminology “nominal matched-dose control” to reflect sequence-length dependency.

3.4 BERTScore Reference-Preference Criterion & Repetition Metric

We define a proxy for clinical factual preference alignment using BERTScore [15] comparisons against reference answer y^+ and hallucinated counter-answer y^- :

$$\text{RefPref}(\hat{y}) = \mathbb{1}[\text{BS}(\hat{y}, y^+) > \text{BS}(\hat{y}, y^-)] \quad (7)$$

where $\text{BS}(\cdot, \cdot)$ denotes BERTScore F1 using `bert-base-multilingual-cased` (layer 9). Cases where $\text{BS}(\hat{y}, y^+) = \text{BS}(\hat{y}, y^-)$ (ties) are counted as non-preferred (evaluated as 0 in binary indicator, remaining in denominator); empirically, exactly 0 out of 500 test questions (0.00%) yielded a tie under greedy decoding. To evaluate text repetition, we record 4-gram repetition (**Rep-4**). For generated sequence $\hat{y}_i$, text is tokenized into word-level tokens using whitespace tokenization (`text.split()`). Let $G_4(\hat{y}_i)$ denote multiset of consecutive 4-grams in $\hat{y}_i$. Sequence-level repetition is defined as:

$$\text{Rep-4}(\hat{y}_i) = \begin{cases} 100 \times \left(1 - \frac{|\text{unique 4-grams in } G_4(\hat{y}_i)|}{|\text{total 4-grams in } G_4(\hat{y}_i)|}\right), & |G_4(\hat{y}_i)| > 0 \\ 0.0\%, & |G_4(\hat{y}_i)| = 0 \end{cases} \quad (8)$$

where short outputs (< 4 tokens, $|G_4(\hat{y}_i)| = 0$) are assigned 0.0%. Performance across $N = 500$ test items is aggregated via sample macro-average mean: $\text{Rep-4} = \frac{1}{N} \sum_{i=1}^N \text{Rep-4}(\hat{y}_i)$.

Evaluator sequence boundaries (`bert-base-multilingual-cased`) enforce maximum limit of 512 subwords. Evaluator statistics across conditions are: (i) **Natural Completion (max_new_tokens=800)**: Baseline outputs averaged 182.1 ± 40.5 mBERT subwords (max 365), Continuous averaged 186.4 ± 44.1 (max 378), Linear Decay averaged 184.2 ± 42.6 (max 372), and Hard Cutoff averaged 184.0 ± 42.1 (max 370). Across all 500 test questions under all 4 conditions, 0 out of 500 outputs (0.00%) exceeded 512-subword limit (0% evaluation truncation), and 0 ties (0.00%) occurred during BERTScore comparison. (ii) **Bounded Stress Testing (max_new_tokens=200)**: Outputs across conditions averaged 142.6 ± 28.3 mBERT subwords (max 200), yielding 0 truncated evaluations (0.00%) and 0 ties (0.00%). RefPref is a semantic reference-preference proxy measuring output proximity to reference vs counter-answers, not a clinician-adjudicated factual correctness rate. Higher RefPref indicates enhanced factual preference relative to distractor without constituting full clinical verification.

3.5 Experimental Setup & Reproducibility Implementation Protocol

Model Checkpoint & Precision. All experiments evaluate Qwen/Qwen2.5-7B-Instruct [4] loaded in 4-bit NormalFloat4 (NF4) quantization (bfloat16 compute dtype via BitsAndBytes).

Chat Template & Verbatim System Prompt. Clinical questions are formatted using Qwen2.5’s native ChatML template via `tokenizer.apply_chat_template(messages, tokenize=False, add_generation_prompt=True)`. System role instruction: “*Bạn là một chuyên gia y được lâm sàng Việt Nam. Hãy trả lời câu hỏi y khoa dưới đây một cách chính xác, khách quan dựa trên Được thư Quốc gia Việt Nam (2nd Edition, 2018).*”.

Exact Hook Location, Insertion & PyTorch Code. Steering hook is attached via PyTorch `register_forward_hook` directly to residual output of decoder block `model.model.layers[8]` (index 8, 9th block of 28-layer architecture). PyTorch hook implementation:

```
class SteeringHookHandler:
    def __init__(self, v_steer, alpha_0=18.0, K=16, schedule="linear_decay"):
        self.v_steer = v_steer
        self.alpha_0 = alpha_0
        self.K = K
        self.schedule = schedule
        self.t = 0

    def reset(self):
        self.t = 0

    def hook_fn(self, module, input, output):
        out_tensor = output[0] if isinstance(output, tuple) else output
        if out_tensor.shape[1] > 1:
            self.t = 0
            return output

        self.t += 1
        if self.schedule == "linear_decay":
            alpha_t = self.alpha_0 * max(0.0, 1.0 - (self.t - 1) / self.K)
        elif self.schedule == "hard_cutoff":
            alpha_t = self.alpha_0 if self.t <= self.K else 0.0
        elif self.schedule == "continuous":
            alpha_t = self.alpha_0
        else:
            alpha_t = 0.0

        if alpha_t != 0.0:
            dev, dt = out_tensor.device, out_tensor.dtype
            v = self.v_steer.to(device=dev, dtype=dt)
            out_tensor[:, -1, :] += alpha_t * v
        return output
```

Final-Token Selection & Vector Extraction. For contrastive vector estimation, hidden state representations $h_s(x_i, y_i)$ are extracted from final token position `([:, -1, :])` of sequence $x_i \oplus y_i$ under teacher-forcing evaluation.

Prefill vs. Decoding, KV Cache & Step Counter t . KV caching (`use_cache=True`) is enabled across all inference runs. During prompt prefill ($t = 0$, sequence length L_{prompt}), hook operates as an identity pass without vector injection. First generated token (y_1) is sampled from unsteered prefill activations. For subsequent single-token autoregressive decoding steps ($t \geq 1$), input sequence length per step is 1 token. Step counter t increments by 1 for each decoding step, modifying position `[:, -1, :]`:

$$\hat{h}_s^{(t)} = h_s^{(t)} + \alpha(t) \cdot v_{\text{steer}}, \quad (9)$$

where $\alpha_0 = 18.0$ across primary experiments. Steering is active during steps $t \in \{1, \dots, K\}$, directly influencing output logits for generated tokens $y_2, y_3, \dots, y_{K+1}$ (K steered tokens in total). Modified activations $\hat{h}_s^{(t)}$ pass to subsequent blocks before being cached in KV memory.

Padding & EOS Termination. Batch generation uses unpadded single-sequence processing (batch size = 1) or left-padding (`padding_side='left'`).

Generation terminates naturally upon emitting official Qwen2.5 end-of-sequence tokens (`<|im_end|>` or `<|endoftext|>`, token IDs 151645/151643) or reaching maximum token cap (`max_new_tokens=200` or `800`).

Open Artifacts & Split Manifest Availability. Dataset manifest (`vietnamese_medical_halueval_15k_specialized.json`), split index files (`train_ids.json`, `val_ids.json`, `test_ids.json`), steering vector tensor weights (`v_steer.pt`), and evaluation scripts are available in public project repository (<https://github.com/anonymous-submission/early-stopping-steering>).

Reproducibility Environment, Compute Precision & Measurement Uncertainty. All experiments were executed on NVIDIA Tesla T4 GPUs (16 GB VRAM, CUDA 12.1) using PyTorch 2.2.0, Hugging Face Transformers 4.38.2, BitsAndBytes 0.42.0, Accelerate 0.27.2, Evaluate 0.4.1, BERTScore 0.3.13 (`bert-base-multilingual-cased`, layer 9), and Rank-BM25 0.2.2. Model weights were loaded in 4-bit NormalFloat4 (NF4) double-quantization with `bfloat16` compute dtype via BitsAndBytes. On Tesla T4 GPUs (Turing architecture, compute capability 7.5 where native BF16 Tensor Cores are unavailable), PyTorch CUDA dispatches operations via automatic hardware fallback to FP16 Tensor Cores (`torch.float16`), maintaining execution speed and precision without numeric instability. All generation runs use deterministic greedy decoding (`temperature=0.0`, `do_sample=False`, `top_p=1.0`). Latency measurements record end-to-end execution time per sample; observed 10 ms difference between steering (23.34 ± 0.28 s) and baseline (23.33 ± 0.30 s) reflects a minimal per-token hook overhead (+0.05 ms per decoding token), substantially smaller than empirical measurement standard deviation (± 0.28 – 0.30 s) across independent GPU profiling sessions. For placebo random vector controls, isotropic Gaussian vectors $v_{\text{rand}} \sim \mathcal{N}(0, I_{3584})$ were sampled using explicit PyTorch seeds (`torch.manual_seed(seed)` for seeds 42–141 for $N = 100$) and normalized to unit ℓ_2 norm $\|v_{\text{rand}}\|_2 = 1.0$.

RAG Benchmark Protocol & Implementation Details. Retrieval is conducted over specialized knowledge base of 14,576 passages extracted from 1,026 monographs of Vietnamese National Drug Formulary (2nd Edition, 2018) [6]. Passages are segmented along monograph section boundaries (mean length 180 ± 45 words / 215 ± 52 tokens). We evaluate three retrieval mechanisms: (i) **Lexical BM25**: Built via `rank_bm25.BM25Okapi` ($k_1 = 1.5$, $b = 0.75$) with Vietnamese word segmentation performed using `pyvi.ViTokenizer`. (ii) **Dense Retrieval**: Embedding representations generated using `BAAI/bge-m3` (1024-dimensional dense vectors) paired with exact cosine similarity search. (iii) **Hybrid RRF**: Combining BM25 and BGE-M3 rank positions using standard RRF ($k_{\text{rrf}} = 60$). Top-1 retrieved passage ($k = 1$) is formatted into prompt context template: “*[Bối cảnh Y tế]: {passage} \n [Câu hỏi]: {question} \n [Trả lời]:*”. Latency measurements cover end-to-end execution on GPU: Retrieval lookup latency (118.4ms BM25, 45.2ms Dense, 142.5ms Hybrid), prompt prefill latency (415.0–440.0ms for mean prompt length 75.4 tokens), and autoregressive decoding latency (6, 750–6, 810ms across $T = 200$ tokens).

4 Results

4.1 Primary Natural Completion Evaluation (`max_new_tokens=800`)

To assess real-world deployable performance where responses complete naturally, we evaluate across test set with `max_new_tokens=800`. Under this constraint,

Table 1. Primary Natural Completion Evaluation ($N_{\text{test}} = 500$, $\text{max_new_tokens}=800$, Greedy Decoding)

Model Condition	RefPref (%)	BERTScore F1	EOS Hit (%)	Rep-4 (%)
Unsteered Baseline	65.40	0.6779	100.00	4.10
Continuous ($K=\infty$)	68.60	0.6705	99.80	4.38
Linear Decay ($K=16$)	67.80	0.6732	99.80	5.37
Hard Cutoff ($K=16$)	70.60	0.6695	100.00	5.35

models achieve an **EOS Hit Rate range of 99.80%–100.00%** (100.00% for Baseline and Hard Cutoff, 99.80% for Continuous and Linear Decay), confirming natural termination without infinite repetition loops.

Table 1 details performance across all four conditions under high-cap natural completion ($\text{max_new_tokens}=800$). Unsteered Baseline and Hard Cutoff achieve 100.00% EOS Hit Rate, while Continuous Steering and Linear Decay achieve 99.80% (499/500). All steering schedules improve reference-preference accuracy over unsteered baseline (65.40%, 327/500): Linear Decay achieves 67.80% (339/500, +2.40 pp), Continuous Steering achieves 68.60% (343/500, +3.20 pp), and Hard Cutoff achieves peak reference-preference accuracy of 70.60% (353/500, +5.20 pp). Under natural completion, higher reference preference is accompanied by a slight decrease in reference BERTScore F1 (0.6779 baseline vs 0.6732 linear decay and 0.6695 hard cutoff) and a moderate increase in 4-gram repetition Rep-4 (4.10% baseline vs 4.38% Continuous, 5.37% Linear Decay, and 5.35% Hard Cutoff). This demonstrates that while early stopping prevents persistent hidden-state norm elevation while maintaining natural termination (EOS Hit 99.80%–100.00%), it does not strictly decrease Rep-4 relative to Continuous steering under natural long completion, illustrating a multi-metric trade-off between preference alignment and lexical diversity.

4.2 Controlled Bounded-Generation Stress Test ($\text{max_new_tokens}=200$)

Under bounded-generation stress testing ($\text{max_new_tokens}=200$), we compare unsteered baseline against early-stopping steering ($\alpha_0 = 18.0$, $K = 16$) across unified $N_{\text{test}} = 500$ test questions. Tables 2 and 3 detail category-wise performance and statistical significance. Early-stopping steering raises reference-preference accuracy across all three categories: Pregnancy Safety (+5.46 pp; 128/165 vs 119/165), Drug Interaction (+3.75 pp; 126/160 vs 120/160), and Special Dosage (+3.43 pp; 132/175 vs 126/175). Overall, reference-preference accuracy increases from 73.00% (365/500) to 77.20% (386/500), a count-derived net improvement of +4.20 percentage points (21/500, 95% Bootstrap CI [+1.40 pp, +7.00 pp]). A two-sided exact binomial McNemar test on discordant pairs ($b = 16$, $c = 37$, net +21 improved queries) confirms statistical significance ($p = 0.0055$; asymptotic $p = 0.0060$). Early-stopping steering also yields a statistically significant paired shift in BERTScore F1 over baseline (Wilcoxon signed-rank $p = 0.017$; paired-bootstrap 95% CI for mean difference [+0.0006, +0.0041]).

4.3 Empirical Activation-Norm Dynamics

To evaluate activation magnitude behavior under continuous vs temporally bounded steering, we define activation norm drift $\Delta_{\text{norm}}^{(t)}$ as mean post-hook ℓ_2

Table 2. BERTScore Reference Preference ($N_{\text{test}} = 500$, $\text{max_new_tokens}=200$)

Category	N	Base (%)	Steer (%)	Δ (pp)
Pregnancy Safety	165	72.12	77.58	+5.46
Drug Interaction	160	75.00	78.75	+3.75
Special Dosage	175	72.00	75.43	+3.43
Overall	500	73.00	77.20	+4.20

Table 3. Paired 2×2 Outcomes for Bounded Stress Test ($N_{\text{test}} = 500$)

	Steered Pref.	Steered Non-Pref.	Total
Base Pref.	349 (<i>a</i>)	16 (<i>b</i>)	365
Base Non-Pref.	37 (<i>c</i>)	98 (<i>d</i>)	135
Total	386	114	500

Table 4. Statistical Significance & Pairwise Contingency Matrix for High-Cap Natural Completion ($N_{\text{test}} = 500$, $\text{max_new_tokens}=800$, Exact Binomial McNemar, $B = 10,000$ Percentile Bootstrap, Seed=42)

Comparison Pair	Base Acc	Steer Acc	Paired (<i>a, b, c, d</i>)	Net Diff	95% Bootstrap CI	McNemar <i>p</i>	Holm <i>p</i> _{adj}
Hard Cutoff ($K=16$) vs Base	65.40%	70.60%	(311, 16, 42, 131)	+5.20 pp	[+2.20 pp, +8.20 pp]	0.00086	0.0026
Continuous ($K=\infty$) vs Base	65.40%	68.60%	(309, 18, 34, 139)	+3.20 pp	[+0.40 pp, +6.00 pp]	0.0365	0.0730
Linear Decay ($K=16$) vs Base	65.40%	67.80%	(308, 19, 31, 142)	+2.40 pp	[-0.40 pp, +5.20 pp]	0.1189	0.1189

magnitude difference relative to unsteered baseline at decoding step t :

$$\Delta_{\text{norm}}^{(t)} = \mathbb{E}_{x \sim \mathcal{D}_{\text{eval}}} \left[\|\hat{h}_8^{(t)}(x)\|_2 - \|h_8^{(t)}(x)\|_2 \right] \quad (10)$$

where $h_8^{(t)}(x)$ denotes unsteered baseline activation vector at Layer 8 for decoding step t , and $\hat{h}_8^{(t)}(x)$ denotes post-hook activation vector after intervention. We tracked post-hook Layer 8 activation ℓ_2 norms ($\|\hat{h}_8^{(t)}\|_2$) across discrete decoding steps $t \in \{1, \dots, 100\}$ ($N = 100$ prompts) under fixed teacher-forcing context. Unsteered baseline generation establishes reference magnitude equilibrium $\|h_8^{(t)}\|_2$ of 53.56 at $t = 10$ and 53.39 ± 0.64 at $t = 50$. Continuous steering ($K = \infty$, $\alpha_0 = 18.0$) induces persistent **norm elevation** ($\Delta_{\text{norm}}^{(50)} = +2.95$), maintaining elevated magnitudes of 56.46 ($t = 10$) and 56.34 ($t = 50$). In contrast, Linear Decay steering ($K = 16$) exhibits **gradual magnitude relaxation**: at $t = 10$ (where decaying coefficient $\alpha(10) = 18.0 \times (1 - 9/16) = 7.875$ is still nonzero), post-hook norm is 54.11—a partial elevation of $\Delta_{\text{norm}}^{(10)} = +0.55$ above baseline 53.56. By $t = 50$ (well past $K = 16$ cutoff), Linear Decay settles at 53.39 ± 0.64 ($\Delta_{\text{norm}}^{(50)} = 0.00$ relative to baseline equilibrium), confirming magnitude relaxation. Hard Cutoff ($K = 16$) produces an abrupt magnitude transition from 56.46 ($t = 10$) back to baseline equilibrium 53.39 ± 0.64 at $t = 50$ ($\Delta_{\text{norm}}^{(50)} = 0.00$). While scalar ℓ_2 norm tracking at Layer 8 directly quantifies magnitude offset relaxation back to baseline levels ($\Delta_{\text{norm}}^{(50)} = 0.00$), scalar norm magnitude tracking at a single intermediate layer demonstrates hook deactivation but does not directly prove complete restoration of multi-layer activation manifold geometry or pinpoint localized truthfulness dynamics across downstream decoder blocks (9...27); multi-dimensional subspace probing remains necessary for complete structural verification.

4.4 Placebo Control Distribution, Manifold Covariance Controls & Real BM25 RAG Results

To test directionality against generic residual perturbation, Table 5 compares Positive Steering ($+v_{\text{steer}}$) against Negative Steering ($-v_{\text{steer}}$), a Multi-Random Placebo Distribution ($N = 100$), and non-oracle Retrieval-Augmented Generation (RAG) baselines based on GPU execution logs on $N_{\text{test}} = 500$.

Intervention along negative steering direction ($-v_{\text{steer}}$, $\alpha_0 = -18.0$) causes a substantial performance drop, reducing reference-preference accuracy to **62.80%** (314/500, 10.20 pp drop below baseline) and BERTScore F1 to 0.6889. Evaluating an expanded distribution of $N = 100$ independent random isotropic vectors ($\{v_{\text{rand}}^{(r)}\}_{r=1}^{100}$, generated via `torch.manual_seed(seed)` for seeds 42–141) yields mean reference-preference accuracy of $55.82\% \pm 4.15\%$ (sample SD, range 46.00%–68.00%). Truthfulness vector $+v_{\text{steer}}$ (77.20%) operates at a **descriptive standardized distance of $+5.15\sigma$ above random vector distribution mean** (one-sided Monte Carlo empirical tail probability $p = 1/101 = 0.0099 < 0.01$, 0 of 100 random directions exceeding $+v_{\text{steer}}$) and **+14.40 pp above negative steering**, demonstrating factual alignment gains are directionally superior over both isotropic Gaussian noise ($+5.15\sigma$) and empirical covariance-matched activation variation ($+2.52\sigma$ / $+2.39$ pp).

To provide a comprehensive retrieval baseline, we evaluate three non-oracle retrieval-augmented configurations across 14,576 Vietnamese National Drug Formulary passages: (i) **BM25 Lexical Retrieval**: Top-1 Recall of **19.20%** (**80.80% Top-1 retrieval miss rate**; Recall@3=28.40%, Recall@5=32.80%, MRR=0.2433). BM25 RAG reference-preference accuracy drops to **68.60%** (343/500, 4.40 pp below baseline; BERTScore F1 0.6970). (ii) **Dense BGE-M3 Retrieval**: Top-1 Recall of **42.60%** (+23.40 pp over BM25) and MRR of **0.5124**, raising reference preference to **74.80%** (374/500). (iii) **Hybrid RRF RAG (BM25+BGE-M3)**: Top-1 Recall of **48.20%** and MRR of **0.5681**, achieving **76.20%** reference preference (381/500; BERTScore F1 0.7190). In contrast, an **Oracle Gold-Context RAG** baseline achieves 89.40% accuracy (447/500) and 0.8002 BERTScore F1.

Table 6 presents paired 2×2 contingency outcome matrices between Linear Decay Steering ($+v_{\text{steer}}$, 386/500, 77.20%) and both retrieval baselines. Exact binomial McNemar testing against Hybrid RRF RAG (381/500, 76.20%) on discordant pairs ($b = 27, c = 22$) yields $p = 0.5682$; no statistically significant difference in reference preference was detected between steering and Hybrid RAG on this benchmark sample. However, Early-Stopping Steering significantly outperforms non-oracle BM25 Lexical RAG (343/500, 68.60%) by **+8.60 pp** (77.20% vs 68.60%; exact binomial McNemar on discordant pairs $b = 59, c = 16$ yields $p = 6.11 \times 10^{-7} < 0.0001$, 95% Bootstrap CI [+5.40 pp, +12.00 pp]) while eliminating retrieval index maintenance, context expansion (35.0 vs 75.4 tokens), and adding negligible computation overhead (+10 ms per 200 tokens, < 0.05%) over baseline.

4.5 Controlled Factorial Ablation Matrix

To evaluate intervention schedules under matched initial steering strength, we conduct a 3×3 factorial ablation holding initial α_0 fixed across Continuous ($K = \infty$), Hard Cutoff ($K = 16$), and Linear Decay ($K = 16$) schedules. A separately labeled Nominal Matched Cumulative D_1 Dose Control row tests the effect of spreading Linear Decay’s total nominal dose D_{decay} uniformly across all $T = 200$ tokens using applied coefficient $\alpha_{\text{match}} = 0.0425\alpha_0$.

Table 7 reports empirical performance alongside unsteered baseline and 95% confidence intervals. Across steering intensities ($\alpha_0 \in \{15, 18, 20\}$), all early-stopping and continuous steering schedules improve reference preference over unsteered baseline (73.00%, 95% CI [68.91%, 76.74%]). At peak intensity

Table 5. Comprehensive Comparison Against Directional Controls, Covariance-Matched Controls ($N = 20$), and Multi-Retriever RAG Baselines ($N_{\text{test}} = 500$, $\text{max_new_tokens}=200$)

Method / Condition	Raw Count	RefPref (%)	Recall@1 (%)	MRR	BERTScore	F1	Latency (s) [†]
Isotropic Gaussian ($N=100$)	—	55.82 ± 4.15	—	—	0.6945		23.33 ± 0.30
Negative Steered ($-v_{\text{steer}}$)	314 / 500	62.80	—	—	0.6889		23.34 ± 0.28
BM25 Lexical RAG	343 / 500	68.60	19.20	0.2433	0.6970		7.32 ± 0.45
Label-Shuffled Steering (v_{shuf})	343 / 500	68.60	—	—	0.6845		23.34 ± 0.28
Unsteered Baseline	365 / 500	73.00	—	—	0.7133		23.33 ± 0.30
Cov-Matched Distribution ($N=20$)	—	74.81 ± 0.95	—	—	0.7182		23.34 ± 0.28
Dense BGE-M3 RAG	374 / 500	74.80	42.60	0.5124	0.7180		7.27 ± 0.42
Hybrid RRF RAG (BM25+BGE-M3)	381 / 500	76.20	48.20	0.5681	0.7190		7.39 ± 0.48
Linear Decay Steered ($+v_{\text{steer}}$)	386 / 500	77.20	—	—	0.7150		23.34 ± 0.28
Oracle Gold-Context RAG	447 / 500	89.40	100.0	1.0000	0.8002		7.17 ± 0.38

[†] Steering-based and RAG-based conditions were profiled in separate Kaggle T4 GPU sessions with different warm-up states; absolute cross-group latency values are not directly comparable. Within-group overhead comparisons are valid: steering adds $\approx +10$ ms per 200 tokens ($+0.05$ ms/token) over baseline (23.34 vs 23.33 s), while RAG retrieval lookup contributes 45–143 ms (45.2 ms Dense, 118.4 ms BM25, 142.5 ms Hybrid) over pure decoding.

Table 6. Paired 2×2 Contingency Outcomes: Linear Decay Steering ($+v_{\text{steer}}$) vs RAG Baselines ($N_{\text{test}} = 500$, $\text{max_new_tokens}=200$)

(A) Steering vs BM25 Lexical RAG ($b = 59, c = 16, p = 6.11 \times 10^{-7}$)			
	BM25 Pref.	BM25 Non-Pref.	Total
Steered Pref.	327	59	386
Steered Non-Pref.	16	98	114
Total	343	157	500
(B) Steering vs Hybrid RRF RAG ($b = 27, c = 22, p = 0.5682$)			
	Hybrid Pref.	Hybrid Non-Pref.	Total
Steered Pref.	359	27	386
Steered Non-Pref.	22	92	114
Total	381	119	500

Table 7. Controlled Factorial Ablation Matrix: Baseline and 3×3 Equal-Initial- α_0 Comparison plus Nominal Matched Dose Control ($N_{\text{test}} = 500$, $\text{max_new_tokens}=200$, greedy decoding). 95% Confidence Intervals (CI) computed via 1,000 bootstrap iterations.

α_0	Intervention Schedule	Applied α_{initial}	RefPref (%)	[95% CI]	ROUGE-L (%)	BERTScore F1	[95% CI]	Rep-4 (%)
–	Unsteered Baseline	0.000	73.00	[68.9, 76.7]	23.12	0.7133	[0.708, 0.718]	3.67
15.0	Continuous ($K=\infty$)	15.000	75.20	[71.2, 78.8]	23.43	0.7136	[0.709, 0.718]	3.65
	Hard Cutoff ($K=16$)	15.000	75.80	[71.8, 79.3]	23.11	0.7146	[0.710, 0.719]	3.73
	Linear Decay ($K=16$)	15.000	75.60	[71.6, 79.2]	23.10	0.7142	[0.709, 0.719]	4.03
	Nominal Matched D_1 Dose Control	0.6375 (α_{match})	75.40	[71.4, 79.0]	23.40	0.7144	[0.709, 0.719]	3.78
18.0	Continuous ($K=\infty$)	18.000	75.60	[71.6, 79.2]	23.36	0.7133	[0.708, 0.718]	3.90
	Hard Cutoff ($K=16$)	18.000	77.00	[73.1, 80.4]	23.16	0.7151	[0.710, 0.720]	3.89
	Linear Decay ($K=16$)	18.000	77.20	[73.3, 80.6]	23.21	0.7150	[0.710, 0.720]	3.74
	Nominal Matched D_1 Dose Control	0.7650 (α_{match})	75.40	[71.4, 79.0]	23.21	0.7140	[0.709, 0.719]	4.38
20.0	Continuous ($K=\infty$)	20.000	75.40	[71.4, 79.0]	23.37	0.7134	[0.709, 0.718]	4.05
	Hard Cutoff ($K=16$)	20.000	76.40	[72.5, 79.9]	23.17	0.7139	[0.709, 0.719]	3.87
	Linear Decay ($K=16$)	20.000	76.60	[72.7, 80.1]	23.15	0.7147	[0.710, 0.719]	3.74
	Nominal Matched D_1 Dose Control	0.8500 (α_{match})	75.20	[71.2, 78.8]	23.33	0.7136	[0.709, 0.718]	4.52

($\alpha_0 = 18$), Linear Decay ($K = 16$) achieves highest reference preference of **77.20%** (95% CI [73.32%, 80.62%]) and BERTScore F1 of **0.7150** (95% CI [0.710, 0.720]), outperforming Continuous Steering (75.60% RefPref, 0.7133 BERTScore F1) while reducing 4-gram repetition Rep-4 to 3.74% compared to Nominal Matched Dose Control (4.38%). At lower intensity ($\alpha_0 = 15$), effect sizes are smaller, with Hard Cutoff achieving 75.80% RefPref and 0.7146 BERTScore F1.

5 Discussion & Limitations

Proxy Nature of BERTScore Criterion. Reference-preference accuracy in Table 2 is a semantic proxy, not a clinician-adjudicated factual correctness rate. Validation against blinded clinician labels is required in future work to establish clinical safety.

Cumulative Dose Confound. Equal initial α_0 produces unequal cumulative intervention doses across schedules (Section 3.3). While Nominal Matched Dose control sets a constant coefficient matching planned nominal dose D_1 under sequence length $T = 200$, actual executed cumulative dose scales with total generated tokens post-prefill ($D_{\text{executed}} = (T_{\text{gen}} - 1)|\alpha_{\text{match}}|$). It does not control for temporal distribution or $D_2 = \sum_t \alpha(t)^2$. We describe this as a controlled comparative ablation rather than causal isolation.

Effect Sizes and Uncertainty. While main reference-preference gains are statistically significant ($p < 0.05$), fine-grained ablation differences between temporal schedules (e.g., a 0.0017 BERTScore difference between Linear Decay and Continuous at $\alpha_0 = 18$) remain small and should be treated as exploratory.

Bootstrap Confidence Interval Methodology. Confidence intervals for reference-preference rate differences are computed via non-parametric percentile bootstrapping ($B = 10,000$ resamples across $N = 500$ test set). In each bootstrap iteration, a sample of size $N = 500$ is drawn with replacement, computing accuracy difference $\Delta^{(b)} = \text{Acc}_{\text{steered}}^{(b)} - \text{Acc}_{\text{baseline}}^{(b)}$. The 95% confidence interval $[L, U]$ corresponds to 2.5th (L) and 97.5th (U) empirical percentiles of $\{\Delta^{(b)}\}_{b=1}^B$.

Hyperparameter Selection Rationale and Validation Protocol. Methodology follows a strict three-stage split workflow: (1) Contrastive vector extraction v_{steer} is performed on training data ($N_{\text{train}} = 10,272$); (2) Layer l , cutoff K , and scale α_0 are tuned exclusively on validation evaluation prompts ($N_{\text{eval}} = 500$ drawn from $N_{\text{val}} = 2,223$); (3) Final statistical evaluation is executed strictly once on held-out test set ($N_{\text{test}} = 500$). Validation sweep results

($N_{\text{eval}} = 500$) are: (i) **Layer Index Sweep** ($l \in \{4, 6, 8, 10, 12, 14\}$): Layer 4 (74.20% RefPref, 0.7042 F1), Layer 6 (76.00%, 0.7105), **Layer 8 (decoder block index 8: 77.40% RefPref [387/500], 0.7155 F1 — Peak)**, Layer 10 (76.80%, 0.7120), Layer 12 (75.40%, 0.7088), Layer 14 (73.80%, 0.7015). Layer 8 yielded peak factual alignment while exhibiting narrow cross-layer score differences ($\Delta \leq 0.0140$ F1). (ii) **Window Threshold Sweep** ($K \in \{4, 8, 12, 16, 24, 32, \infty\}$): $K = 4$ (74.80%), $K = 8$ (76.20%), $K = 12$ (76.80%), $K = 16$ (**77.40% RefPref [387/500] — Peak**), $K = 24$ (76.60%, norm drift +1.12), $K = 32$ (76.00%, norm drift +1.98), $K = \infty$ Continuous (75.80%, norm drift +2.95). Vector injection is active during initial steps ($t \in [1, 16]$); continuing steering past $t = 16$ produced diminishing factual returns while accumulating norm drift (+2.95). (iii) **Scaling Multiplier Sweep** ($\alpha_0 \in \{5, 10, 15, 18, 20, 25\}$): $\alpha_0 = 5$ (74.40%), $\alpha_0 = 10$ (75.80%), $\alpha_0 = 15$ (76.40%), $\alpha_0 = 18$ (**77.40% RefPref [387/500] — Peak**), $\alpha_0 = 20$ (76.80%), $\alpha_0 = 25$ (72.20%, onset of quality degradation). The held-out test set ($N_{\text{test}} = 500$) subsequently yielded 77.20% (386/500) under Linear Decay ($K = 16, T = 200$) and 70.60% (353/500) under Hard Cutoff ($K = 16$, natural completion).

Natural Termination Validation. Extending generation cap to `max_new_tokens=800` confirms natural completion achieves 99.80%–100.00% EOS hit rate range across conditions (100.00% Hard Cutoff and Baseline, 99.80% Continuous and Linear Decay) without text repetition loops. A statistically significant preference improvement (+5.20 pp for Hard Cutoff, 70.60% vs 65.40% baseline; exact binomial McNemar $p = 0.00086$, Holm-Bonferroni adjusted $p_{\text{adj}} = 0.0026$, 95% Bootstrap CI [+2.20 pp, +8.20 pp]) was confirmed under natural long completion.

Layer-Wise Subspace Probing. To validate layer selection, we swept activation steering across even-indexed layers $l \in \{4, 6, 8, 10, 12, 14\}$ on validation subset ($N_{\text{eval}} = 500$). BERTScore F1 varied within a narrow band (0.7015–0.7155, $\Delta \leq 0.0140$), suggesting factual preference interventions are accessible across multiple intermediate decoder layers (4...14) rather than strictly restricted to a single layer. Layer 8 was selected based on peak validation performance; narrow cross-layer score difference ($\Delta \leq 0.0140$) confirms framework robustness to moderate layer selection shifts, though formal causal decomposition of representation localization warrants future work.

Generalizability. Experiments focus on Qwen2.5-7B-Instruct [4] with Vietnamese medical data. The framework—contrastive vector estimation followed by decayed inference-time intervention—is conceptually applicable to other residual-stream transformers, subject to architecture-specific validation.

6 Conclusion

We introduced Early-Stopping Activation Steering, a temporally bounded inference-time intervention for Vietnamese clinical Small Language Models. Using the ViHaluEval-Medical benchmark (14,674 records derived from Vietnamese National Drug Formulary, 2nd Edition, 2018), we demonstrated restricting activation steering to the first $K = 16$ decoding steps ($t \in [1, 16]$), influencing output tokens $y_2 \dots y_{17}$ yields reference-preference improvements with zero model-weight updates and negligible generation latency overhead (+10 ms per 200 tokens). Under primary high-cap natural completion (`max_new_tokens=800`), steering achieves EOS hit rate range of 99.80%–100.00% across conditions

(100.00% Hard Cutoff and Baseline, 99.80% Continuous and Linear Decay) while raising reference preference from 65.40% (327/500) to 67.80% (339/500) for Linear Decay ($p = 0.1189$, exploratory) and 70.60% (353/500) for Hard Cutoff ($p = 0.00086$, $p_{\text{adj}} = 0.0026$), exhibiting a Pareto trade-off with reference BERTScore F1 and 4-gram repetition. Under bounded stress testing (`max_new_tokens=200`), early-stopping steering raises reference-preference accuracy from 73.00% (365/500) to 77.20% (386/500) (+4.20 pp, 95% Bootstrap CI [+1.40 pp, +7.00 pp], exact binomial McNemar $p = 0.0055$, $b = 16$, $c = 37$). Empirical negative steering along $-v_{\text{steer}}$ (62.80% accuracy) and placebo control distribution ($N = 100$, $55.82\% \pm 4.15\%$ SD) confirm $+v_{\text{steer}}$ operates at a descriptive $+5.15\sigma$ distance above random perturbation ($p = 1/101 = 0.0099 < 0.01$), while comparison against real non-oracle BM25 RAG (68.60%, 75.4 prompt tokens) highlights steering’s zero-retrieval latency, zero-context-memory advantage, and superior reference-preference rate (+8.60 pp, $p = 6.11 \times 10^{-7}$). Requiring zero model retraining, early-stopping steering offers a practical alignment mechanism warranting further clinician-adjudicated validation.

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References

1. Zou, A. et al.: Representation engineering: A top-down approach to AI transparency. arXiv preprint arXiv:2310.01405 (2023)
2. Li, K. et al.: Inference-time intervention: Eliciting truthful answers from a language model. In: Proc. NeurIPS, vol. 36, pp. 41451–41530 (2023)
3. Singhal, K. et al.: Large language models encode clinical knowledge. *Nature* **620**, 172–180 (2023)
4. Qwen Team: Qwen2.5 Technical Report. arXiv preprint arXiv:2412.15115 (2024)
5. Lewis, P. et al.: Retrieval-augmented generation for knowledge-intensive NLP tasks. In: Proc. NeurIPS, vol. 33, pp. 9459–9474 (2020)
6. Ministry of Health of Vietnam: Vietnamese National Drug Formulary (Dược thư Quốc gia Việt Nam). 2nd edn. Medical Publishing House, Hanoi (2018)
7. Ji, Z. et al.: Survey of hallucination in natural language generation. *ACM Computing Surveys* **55**(12), 1–38 (2023)
8. Turner, A.M. et al.: Steering language models with activation engineering. arXiv preprint arXiv:2308.10248 (2023)
9. Rinsky, N. et al.: Steering Llama 2 via contrastive activation addition. In: Proc. ACL, pp. 15504–15522 (2024)
10. Hu, E.J. et al.: LoRA: Low-rank adaptation of large language models. In: Proc. ICLR (2022)
11. Yoran, S. et al.: Making retrieval-augmented language models robust to irrelevant context. In: Proc. ICLR (2024)
12. Kirkpatrick, J. et al.: Overcoming catastrophic forgetting in neural networks. *Proc. PNAS* **114**(13), 3521–3526 (2017)
13. Li, J. et al.: HaluEval: A large-scale hallucination evaluation benchmark for large language models. In: Proc. EMNLP, pp. 6449–6464 (2023)
14. Lin, C.-Y.: ROUGE: A package for automatic evaluation of summaries. In: Text Summarization Branches Out, pp. 74–81 (2004)

15. Zhang, T. et al.: BERTScore: Evaluating text generation with BERT. In: Proc. ICLR (2020)
16. Lin, S. et al.: TruthfulQA: Measuring how models mimic human falsehoods. In: Proc. ACL, pp. 3214–3252 (2022)